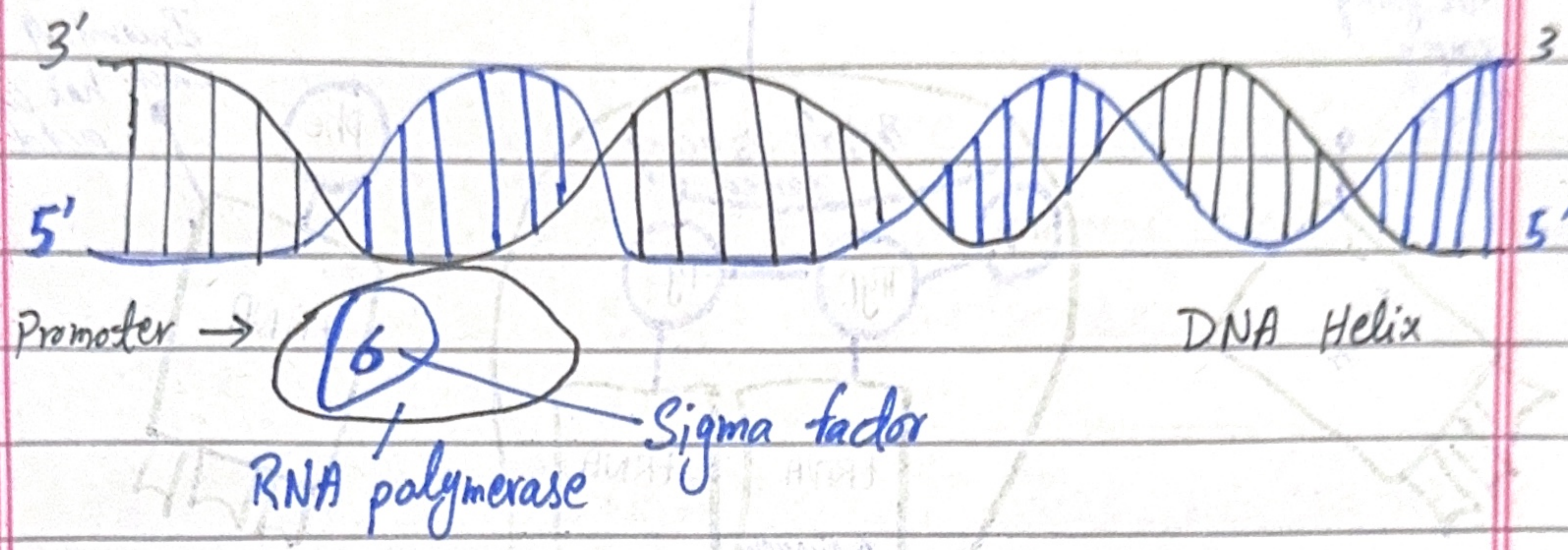
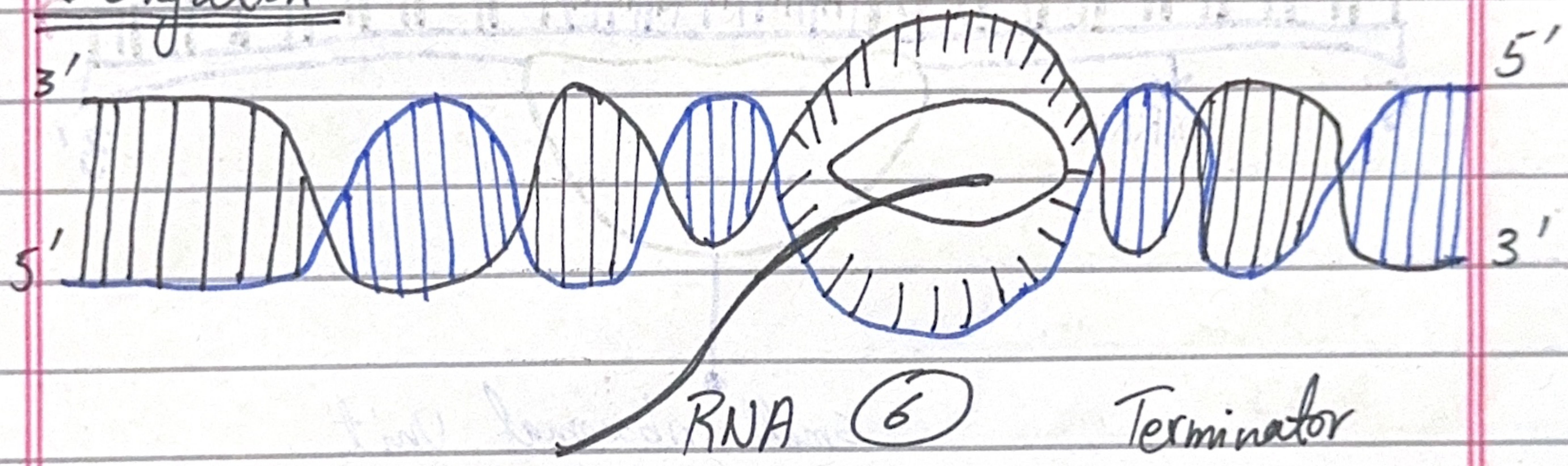
FA24-BSE-042 Assignment #3 M.Abdullah.

1) A Labelled Diagram showing the process of transcription.

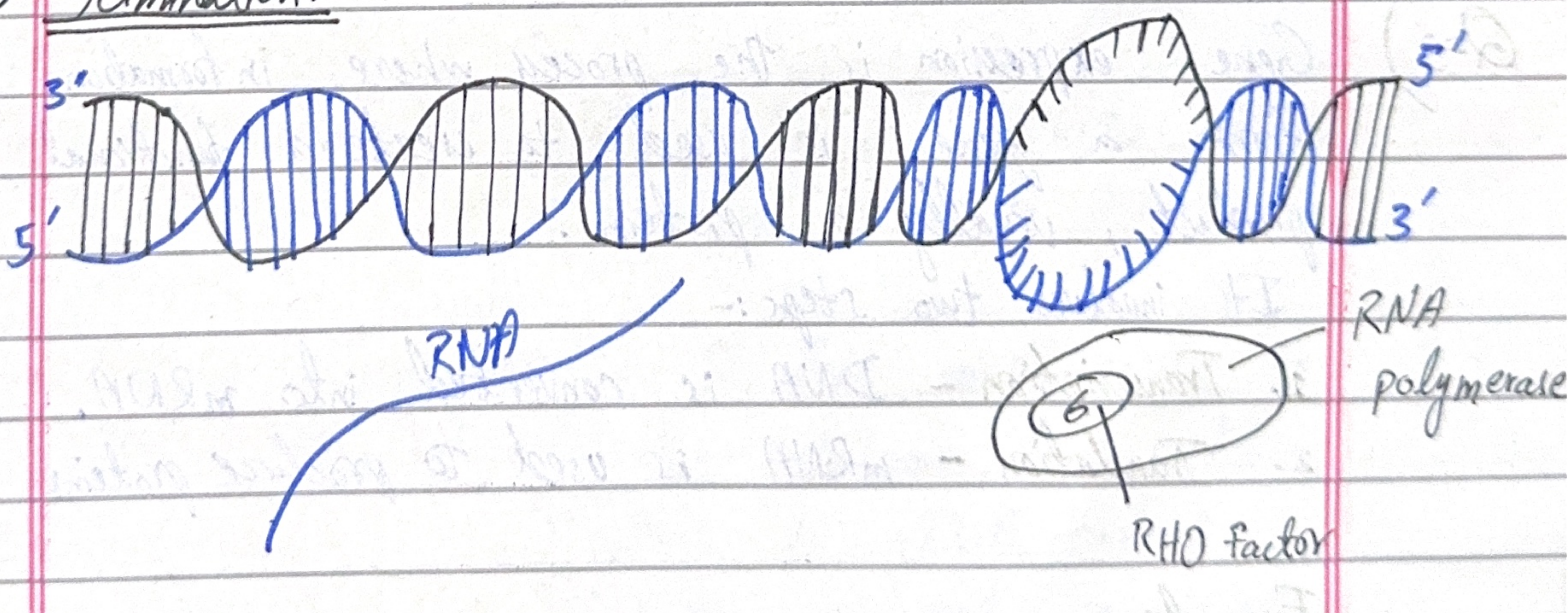
⇒ Initiation:-



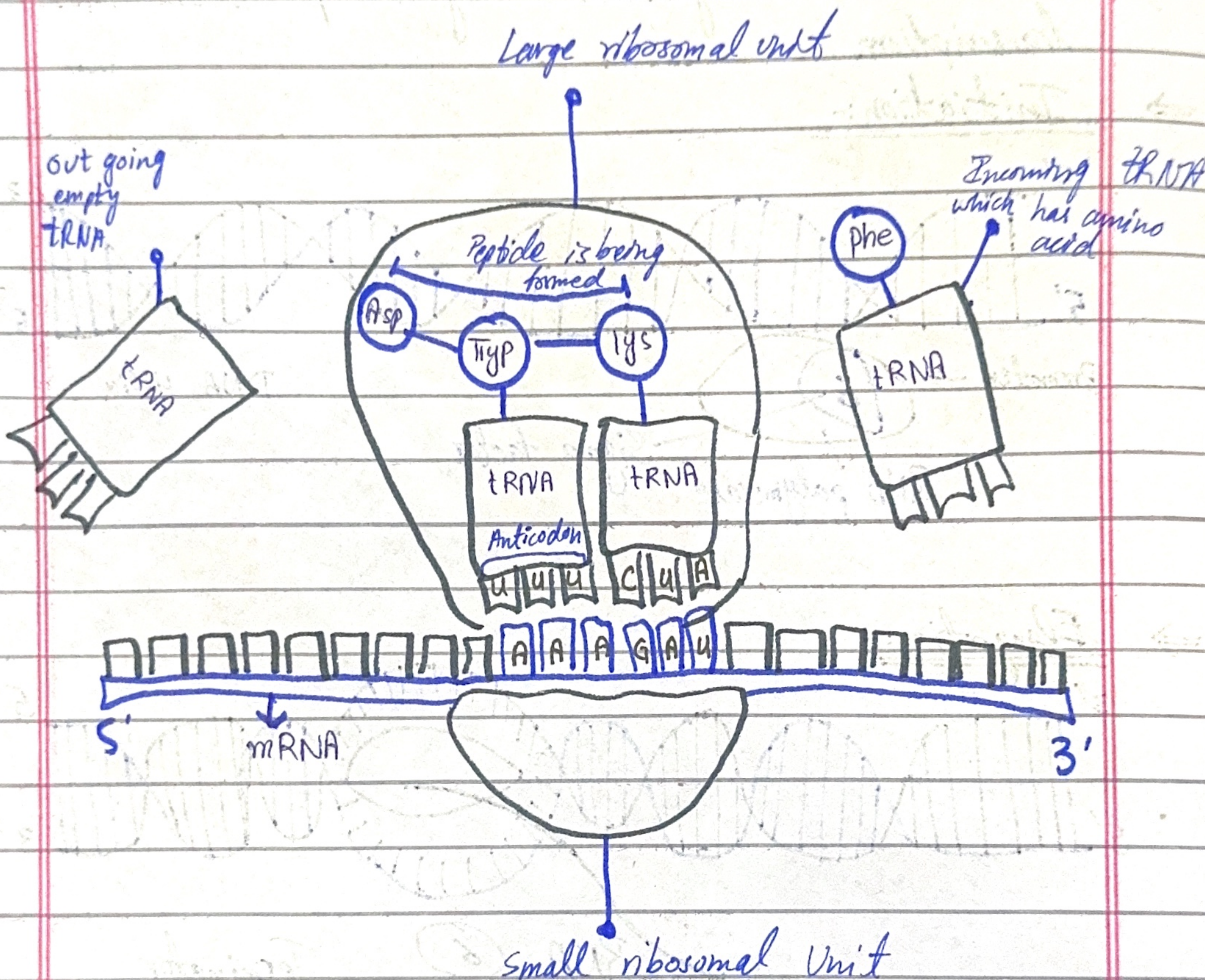
⇒ Elongation:-



⇒ Termination:-



Q=2) Labelled Diagram of Translation.



Q=3) Gene expression is the process where information from a gene is used to create a functional product, usually a protein.

It involves two steps:-

1. Transcription - DNA is converted into mRNA.
2. Translation - mRNA is used to produce proteins.

Example:-

The transcript describes how the DNA sequence is transcribed into mRNA and, then translated into a protein. If a mutation occurs (e.g, changing a codon from coding for arginine to coding for glutamine), the protein may fold differently, affecting its function.

Q-4) What is codon? What is the difference between a codon and an anti-codon?

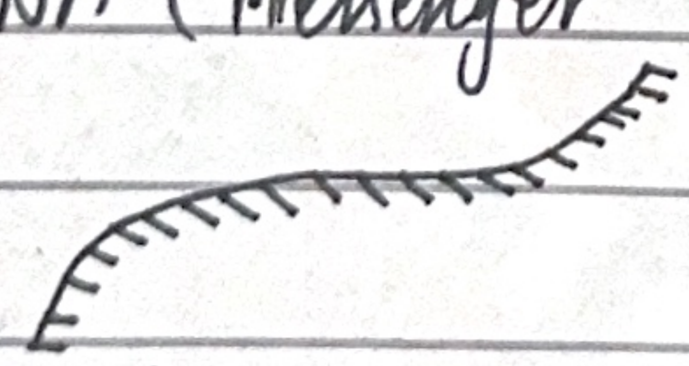
⇒ Codon : A sequence of three nucleotides on m-RNA that codes for an amino acid (Example : AUG codes for methionine).

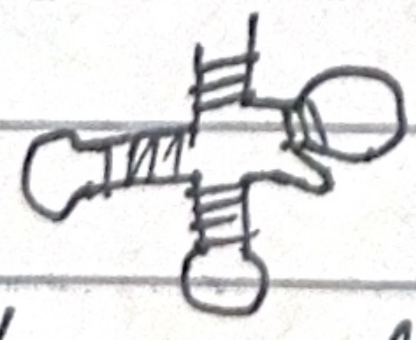
⇒ Anti-Codon : A sequence of three nucleotides on tRNA that is complementary to the codon. (Example: If the codon is AUG, the anticodon is UAC.)

Key difference:-

Codon is found on mRNA
Anti-codon is found on t-RNA

Q-5) There are 3 types of RNA :-

1) mRNA (Messenger RNA) - Carries genetic information from DNA to the ribosome
⇒ 

2) tRNA (Transfer RNA) - Transfers amino acids to the ribosome and matches them to mRNA codons.
⇒ 

3) rRNA (Ribosomal RNA) - A structural component of ribosomes, where protein synthesis occurs.
⇒